

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling. (BL4)

QUESTIONS: 05

Total Marks:100

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I A.Sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

RUBRICS FOR CALCULATION

TECHNICAL PRESENTATION					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Technical Presentation

Choose ANY ONE Scenario:

1. Student Performance Analysis System

A college wants to develop a Python-based system to analyze student marks, attendance, and pass percentage. Explain how:

- A. Python programming basics are used
- B. Pandas and NumPy are used for data handling
- C. Matplotlib and Seaborn are used for result visualization

2. Disease Prediction System

A hospital is planning to build a simple disease prediction system using patient details such as age, blood pressure, sugar level, and cholesterol. Explain how:

- A. Python libraries support medical data analysis
- B. Scikit-Learn is used for model training and prediction
- C. Jupyter Notebook or Google Colab helps in implementation

3. Sales Forecasting Dashboard

A retail company wants to analyze monthly sales data and predict future sales trends. Explain how:

- A. Pandas is used for reading and cleaning sales data
- B. SciPy and Scikit-Learn are used for analysis and prediction
- C. Matplotlib and Seaborn are used to create business visualizations

4. Image Classification Application

A startup is developing an image classification application to identify food items from images. Explain how:

- A. Python is used as the development language
- B. TensorFlow and PyTorch are used for deep learning model development
- C. Notebook environments help in training, testing, and visualizing results

5. Smart Agriculture Data Monitoring

An agriculture research team collects temperature, humidity, soil moisture, and crop growth data from sensors. Explain how:

- A. NumPy is used for numerical computation
- B. Pandas is used for organizing sensor data
- C. Scikit-Learn and visualization libraries support crop condition analysis

2. Disease Prediction System

A hospital is planning to build a simple disease prediction system using patient details such as age, blood pressure, sugar level, and cholesterol. Explain how:

DataSet:

```
Age,BloodPressure,Sugar,Cholesterol,Disease
25,120,90,180,0
45,150,180,250,1
35,130,110,200,0
50,160,200,270,1
60,170,210,280,1
30,125,95,190,0
55,155,190,260,1
40,140,130,220,0
28,118,88,175,0
65,175,220,290,1
```

A.Python libraries support medical data analysis

```
import pandas as pd
from tkinter import Tk
from tkinter.filedialog import askopenfilename
```

```
Tk().withdraw()
```

```
print("Please upload the disease dataset..")
```

```
file_path = askopenfilename(
    title="Upload Disease Dataset",
    filetypes=[("CSV Files", "*.csv")]
)
```

```
if file_path == "":
    print("No file selected.")
    exit()
```

```
df = pd.read_csv(file_path)
```

```
print("\nDataset Loaded Successfully!\n")
print(df)
```

```
print("\nFirst 5 Records")
print(df.head())
```

```
print("\nDataset Information")
print(df.info())
```

```
print("\nStatistical Summary")
print(df.describe())
```

Please upload the disease dataset...

Dataset Loaded Successfully!

	Age	BloodPressure	Sugar	Cholesterol	Disease
0	25	120	90	180	0
1	45	150	180	250	1
2	35	130	110	200	0
3	50	160	200	270	1
4	60	170	210	280	1
5	30	125	95	190	0
6	55	155	190	260	1
7	40	140	130	220	0
8	28	118	88	175	0
9	65	175	220	290	1

First 5 Records

	Age	BloodPressure	Sugar	Cholesterol	Disease
0	25	120	90	180	0
1	45	150	180	250	1
2	35	130	110	200	0
3	50	160	200	270	1
4	60	170	210	280	1

Dataset Information

<class 'pandas.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	Age	10 non-null	int64
1	BloodPressure	10 non-null	int64
2	Sugar	10 non-null	int64
3	Cholesterol	10 non-null	int64
4	Disease	10 non-null	int64

dtypes: int64(5)

memory usage: 532.0 bytes

None

Statistical Summary

B. Scikit-Learn is used for model training and prediction

```
import pandas as pd
from tkinter import Tk
from tkinter.filedialog import askopenfilename
from sklearn.tree import DecisionTreeClassifier
```

```
Tk().withdraw()
```

```
file = askopenfilename(title="Select Disease Dataset", filetypes=[("CSV Files", "*.csv")])
```

```
if not file:
    print("No file selected")
    exit()
```

```
df = pd.read_csv(file)
```

```
X = df[["Age", "BloodPressure", "Sugar", "Cholesterol"]]
y = df["Disease"]
```

```
model = DecisionTreeClassifier(random_state=42)
model.fit(X, y)
```

```
age = int(input("Enter Age: "))
bp = int(input("Enter Blood Pressure: "))
sugar = int(input("Enter Sugar Level: "))
chol = int(input("Enter Cholesterol: "))
```

```
new_data = pd.DataFrame({
    "Age": [age],
    "BloodPressure": [bp],
    "Sugar": [sugar],
    "Cholesterol": [chol]
})
```

```
prediction = model.predict(new_data)
```

```
if prediction[0] == 1:
    print("Prediction: Disease Detected")
else:
    print("Prediction: No Disease")
```

```
===== RESTART: C:/Users/USER/OneDrive/Documents/FODS ASSESSMENTS/CO2/CO2_AT1 (2).py =====
Enter Age: 19
Enter Blood Pressure: 160
Enter Sugar Level: 200
Enter Cholesterol: 280
Prediction: No Disease

===== RESTART: C:/Users/USER/OneDrive/Documents/FODS ASSESSMENTS/CO2/CO2_AT1 (2).py =====
Enter Age: 55
Enter Blood Pressure: 160
Enter Sugar Level: 200
Enter Cholesterol: 270
Prediction: Disease Detected
|
```

C. Jupyter Notebook or Google Colab helps in implementation

```
import pandas as pd
from tkinter import Tk
from tkinter.filedialog import askopenfilename
from sklearn.tree import DecisionTreeClassifier
```

```

Tk().withdraw()
file = askopenfilename(title="Select Disease Dataset", filetypes=[("CSV Files", "*.csv")])

if not file:
    print("No file selected")
    exit()

df = pd.read_csv(file)

X = df[["Age", "BloodPressure", "Sugar", "Cholesterol"]]
y = df["Disease"]

model = DecisionTreeClassifier(random_state=42)
model.fit(X, y)

print("Model Trained Successfully")

age = int(input("Enter Age: "))
bp = int(input("Enter Blood Pressure: "))
sugar = int(input("Enter Sugar Level: "))
chol = int(input("Enter Cholesterol: "))

test = pd.DataFrame({
    "Age": [age],
    "BloodPressure": [bp],
    "Sugar": [sugar],
    "Cholesterol": [chol]
})

result = model.predict(test)

if result[0] == 1:
    print("Disease Detected")
else:
    print("No Disease")

```

```

===== RESTART: C:/Python37
Model Trained Successfully
Enter Age: 60
Enter Blood Pressure: 170
Enter Sugar Level: 210
Enter Cholesterol: 280
Disease Detected
|

```